The Long Noncoding RNA *RMST* Interacts with SOX2 to Regulate Neurogenesis

Shi-Yan Ng,[1,2] Gireesh K. Bogu,[1,5] Boon Seng Soh,[2] and Lawrence W. Stanton[1,3,4,*]
[1]Stem Cell and Developmental Biology Group, Genome Institute of Singapore, 60 Biopolis Street, Singapore 138672, Singapore
[2]Stem Cell and Regenerative Biology, Harvard University, Cambridge, MA 02138, USA
[3]Department of Biological Sciences, National University of Singapore, Singapore 117543, Singapore
[4]School of Biological Sciences, Nanyang Technological University, Singapore 637551, Singapore
[5]Present address: Centre for Genomic Regulation, 08003 Barcelona, Spain
*Correspondence: stantonl@gis.a-star.edu.sg
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SUMMARY

Long noncoding RNAs (lncRNAs) are abundant in the mammalian transcriptome, and many are specifically expressed in the brain. We have identified a group of lncRNAs, including rhabdomyosarcoma 2-associated transcript (*RMST*), which are indispensable for neurogenesis. Here, we provide mechanistic insight into the role of human *RMST* in modulating neurogenesis. *RMST* expression is specific to the brain, regulated by the transcriptional repressor REST, and increases during neuronal differentiation, indicating a role in neurogenesis. *RMST* physically interacts with SOX2, a transcription factor known to regulate neural fate. *RMST* and SOX2 coregulate a large pool of downstream genes implicated in neurogenesis. Through RNA interference and genome-wide SOX2 binding studies, we found that *RMST* is required for the binding of SOX2 to promoter regions of neurogenic transcription factors. These results establish the role of *RMST* as a transcriptional coregulator of SOX2 and a key player in the regulation of neural stem cell fate.

INTRODUCTION

Long noncoding RNAs (lncRNAs) are emerging as key players in regulating developmental processes. Loss-of-function studies carried out in embryonic stem cells (ESCs) and somatic progenitor cells revealed that differentiation pathways are defective upon lncRNA knockdown (Guttman et al., 2011; Hu et al., 2011; Klattenhoff et al., 2013; Ng et al., 2012; Sun et al., 2013), indicating that lncRNAs have important cellular functions. The mechanisms of some of these lncRNAs have been determined at the molecular level. In pluripotent stem cells, lncRNAs associate with chromatin-modifying complexes and transcription factors to maintain stemness (Guttman et al., 2011; Mercer et al., 2008; Ng et al., 2012), whereas a number of lncRNAs act in *cis* to regulate messenger RNA (mRNA) expression during development (Alfano et al., 2005; Feng et al., 2006; Onoguchi et al., 2012; Tochitani and Hayashizaki, 2008).

lncRNAs also have important roles in modulating neural cell-fate decisions (Mercer et al., 2008; Mercer et al., 2010; Ponjavic et al., 2009). Using a genome-wide approach to screen for neurogenic lncRNAs, we previously identified rhabdomyosarcoma 2-associated transcript (*RMST*) as an lncRNA required for neuronal differentiation (Ng et al., 2012). Recent studies in mouse models have uncovered a plausible role of *Rmst* in the brain. The mouse *Rmst* was found to be highly expressed in midbrain dopaminergic neuronal precursors in comparison to the rest of the brain, and these studies also showed that *Rmst* coexpressed with the midbrain transcription factor Lmx1a in the developing mouse brain (Uhde et al., 2010).

Although previous studies presented evidence suggesting that *RMST* has a function in brain development, little is known about the underlying mechanisms in which this lncRNA functions. Unlike many nuclear lncRNAs, *RMST* does not associate with the Polycomb repressive complex 2 or the REST-coREST complex (Ng et al., 2012). In this study, we elucidate a role of *RMST* in modulating neurogenesis. *RMST* associates with SOX2, one of the most important transcription factors controlling neural stem cell (NSC) fate. By means of genome-wide chromatin immunoprecipitation (ChIP) and chromatin-RNA association analyses, we show that *RMST* behaves as a transcriptional coregulator of SOX2, and that this SOX2-*RMST* complex regulates many common target genes. The cobinding of the protein-RNA complex to promoter regions of neurogenic genes is important for neuronal gene activation.

RESULTS

RMST Is a Transcriptionally Regulated, Alternatively Spliced lncRNA Gene

RMST was first identified as an lncRNA important for neuronal specification in a previous study that found *RMST* to be strongly upregulated upon neuronal differentiation of human ESCs (hESCs) (Ng et al., 2012). Knockdown of *RMST* also prevented neuronal differentiation, prompting additional studies to reveal the mechanistic role of *RMST* in regulating neurogenesis. One of the hallmarks of a functional lncRNA is its specific and regulated expression (Amaral and Mattick, 2008). *RMST* (also known as *NCRMS*) is transcriptionally regulated by PAX2, a transcription factor required for the formation of the isthmic organizer at the midbrain-hindbrain boundary (Bouchard et al., 2005). In addition, genome-wide ENCODE transcription factor binding

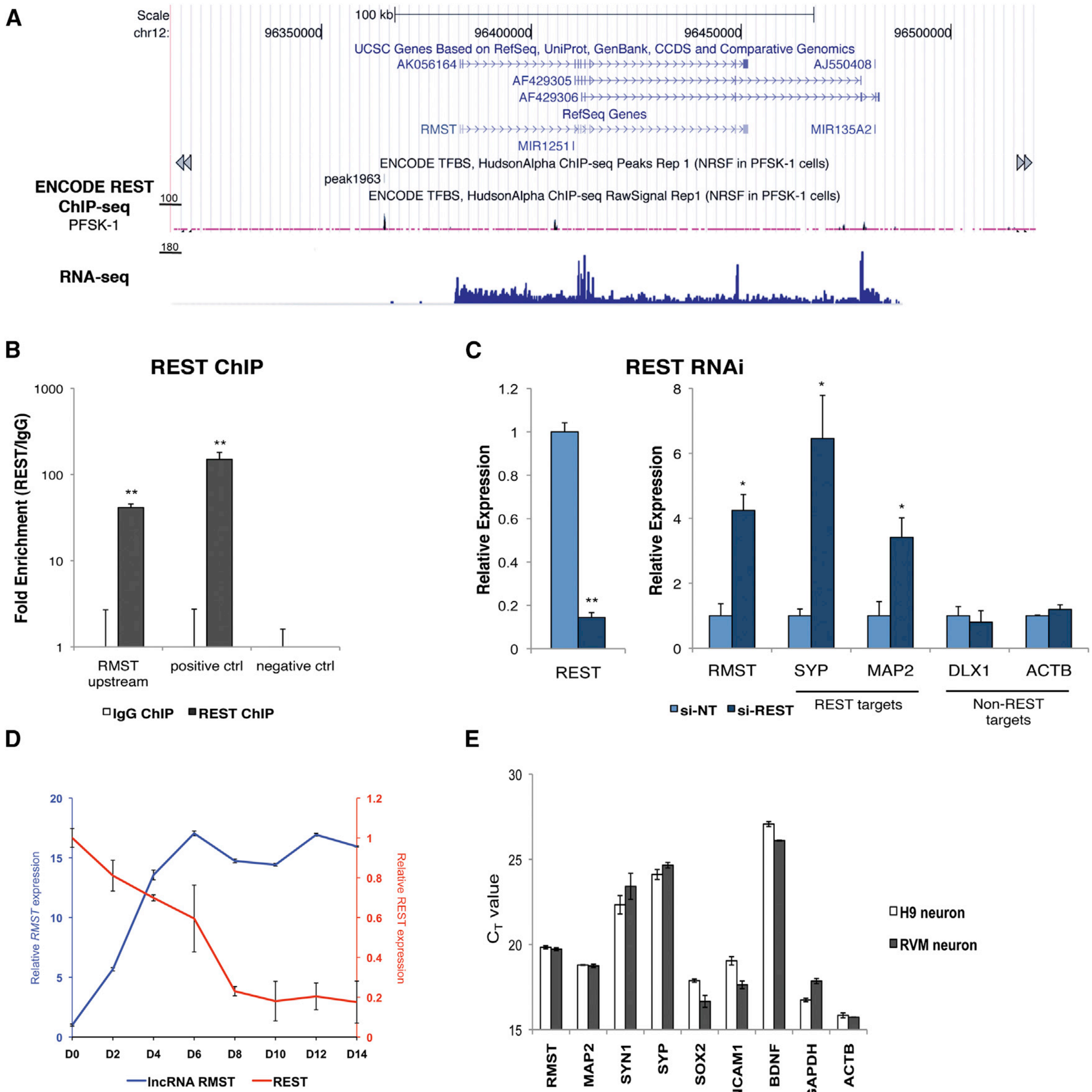


Figure 1. The lncRNA *RMST* Is Regulated by the Transcription Factor REST and Upregulated during Neuronal Differentiation

(A) Human *RMST* is transcribed from chr12 and is alternatively spliced into three isoforms: *AK056164*, *AF429305*, and *AF429306*. From the ENCODE REST (NRSF) ChIP-seq tracks, a binding event was observed upstream of the *RMST* start sites. An RNA-seq track of *RMST* expression in cultured neurons is also shown.

(B) REST ChIP was performed in the human NSC line ReN-VM, and the enrichment of REST at specific genomic locations was measured by qPCR. Error bars are SD. *p < 0.05; **p < 0.01.

(C) qPCR indicated that the knockdown of REST was efficient (>80%; p = 0.0015) 48 hr after transfection of siRNAs and that, upon REST knockdown, expression of *RMST* increased (p = 0.0174) along with known targets of REST. Expression levels are normalized to the nontarget siRNA control (si-NT). Error bars are SD. *p < 0.05; **p < 0.01.

(D) The dynamic changes of *RMST* and *REST* transcript levels during differentiation of ReN-VM cells were tracked over a 14-day period, showing that *RMST* and REST exhibited reciprocal expression in differentiating ReN-VM cells. Expression levels are normalized to day 0 values. Error bars are SD.

(E) *RMST* is an abundant transcript in cultured neurons from the ReN-VM cell line (RVM neuron) as well as from H9-derived neurons (H9 neuron). qPCR was performed, and it compared the expression levels of *RMST* and neuronal transcripts.

See also Figure S1.

analyses also revealed a REST binding site upstream of *RMST*, suggesting that the lncRNA is transcriptionally regulated by REST (Figure 1A). REST (or neuron-restrictive silencing factor [NRSF]) is known as a master negative regulator that controls neurogenesis (Abrajano et al., 2009; Gao et al., 2011) by dissociating from RE1 sites in order to trigger the activation of neuronal genes (Ballas et al., 2005).

To confirm REST regulation of *RMST*, we performed ChIP experiments and gene expression analysis upon REST knockdown in the human NSC line ReN-VM. Using a monoclonal

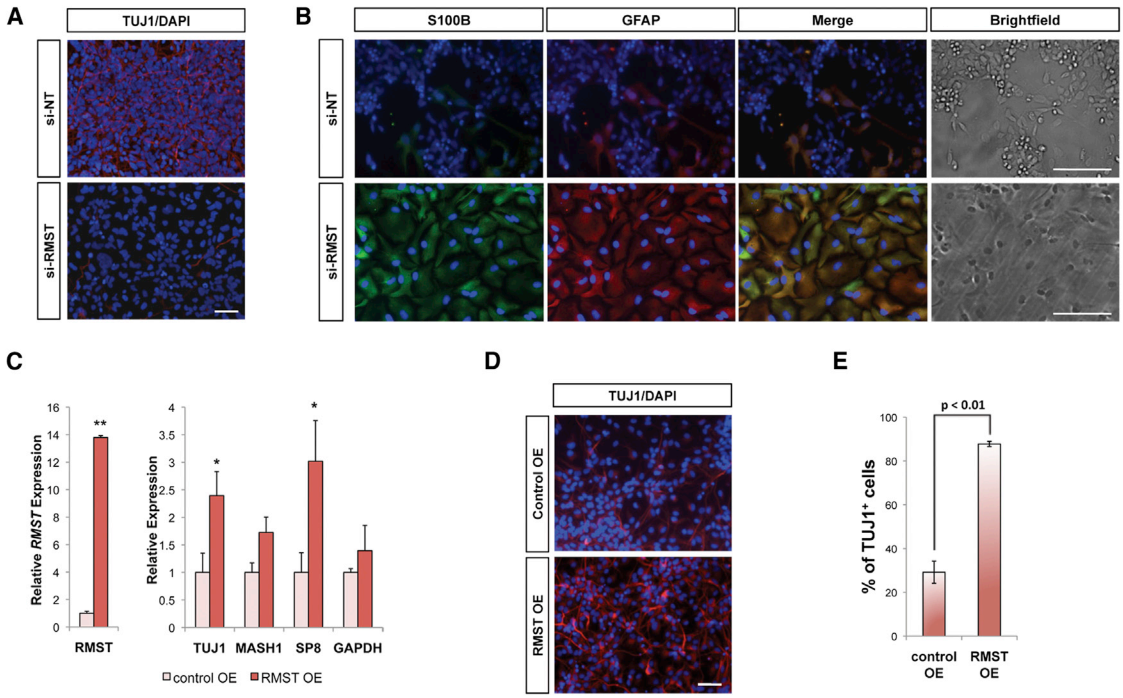


Figure 2. *RMST* Is Necessary for Neuronal Differentiation

(A) Knockdown of *RMST* in H9-derived neural progenitor cells during N2B27-mediated differentiation led to a loss of TUJ1-expressing neurons.

(B) Knockdown of RMST (si-RMST) during N2B27-mediated differentiation of H9-derived NPCs resulted in most cells gaining an astrocyte phenotype (S100B and GFAP coexpression), whereas, in cultures that received the nontarget siRNA (si-NT), only a small proportion of cells coexpressed S100B and GFAP.

(C) Enforced expression of *RMST* to physiological levels resulted in the upregulation of neuronal markers *TUJ1*, *MASH1*, and *SP8* 48 hr after transfection. Error bars are SD. *$p < 0.05$; **$p < 0.01$.

(D and E) Neuronal induction was measured 7 days after transfection of a control RNA or the *RMST* RNA by an abundance of TUJ1-expressing neurons with immunofluorescence and flow cytometry, respectively. Representative images from each treatment condition are shown. Overexpression of *RMST* led to a 3-fold increase in TUJ1-expressing neurons ($p = 0.0039$). Error bars are SD. Scale bars represent 100 μm.

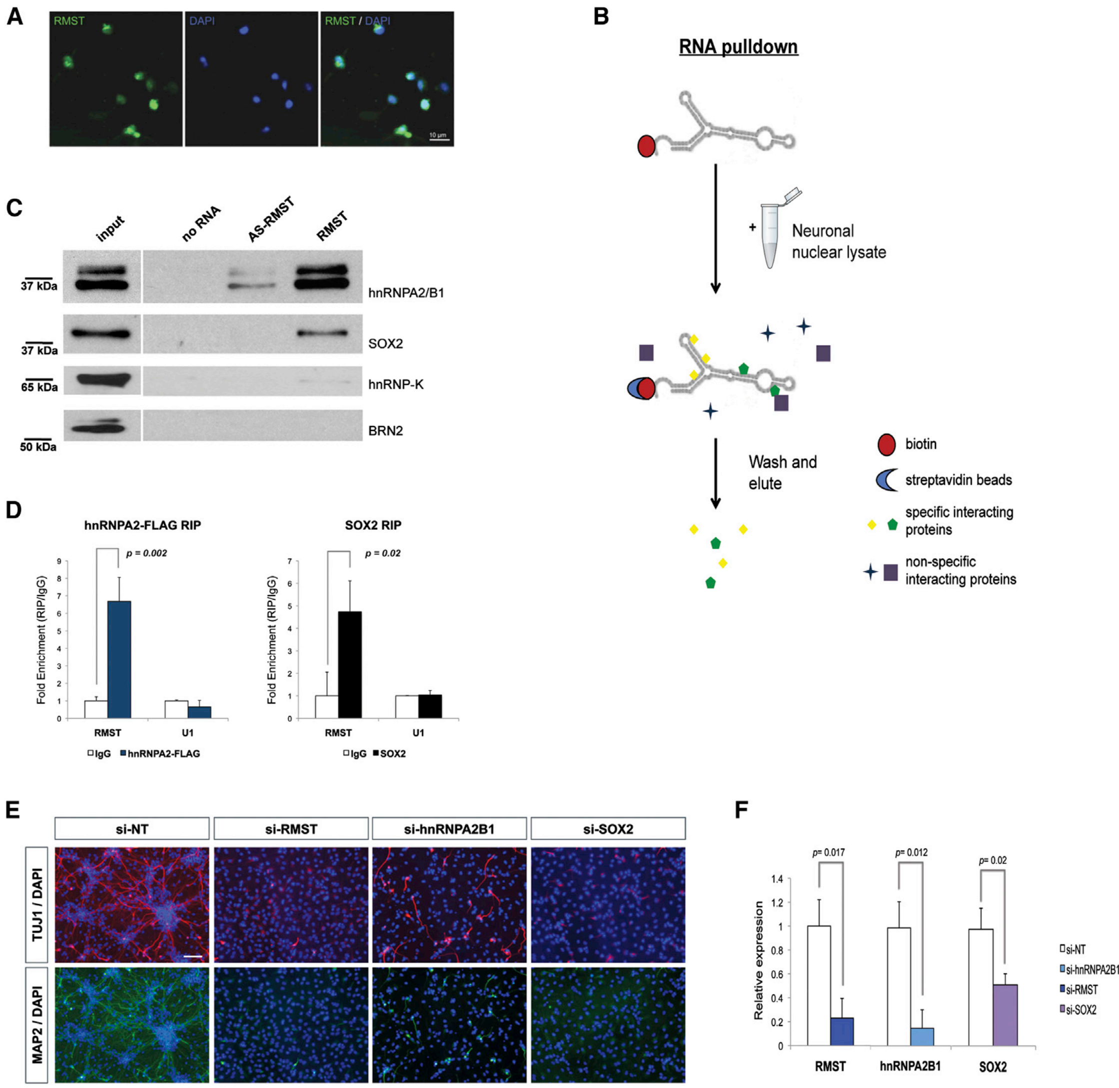
See also Figure S4.

REST antibody to isolate REST-associated chromatin fragments, we verified that REST binds upstream of the lncRNA gene *RMST* (Figure 1B). In addition, knockdown of REST by small interfering RNAs (siRNAs) resulted in the upregulation of *RMST*, confirming that *RMST* is transcriptionally regulated by REST (Figure 1C). In neuronal differentiation, REST expression level decreases, and we expected a reciprocal increase of *RMST* expression as REST levels declined. To test this prediction, we measured the expression levels of *RMST* and REST over a 14-day differentiation of the ReN-VM NSCs. The results clearly show increased *RMST* expression in conjunction with reduced REST levels during neurogenesis (Figure 1D). Therefore, as a REST-regulated lncRNA, *RMST* constitutes part of the extended REST regulatory network of macroRNAs that play a role in nervous system development (Johnson et al., 2009).

RMST is a multiexon lncRNA that gives rise to three alternatively spliced isoforms in human cells: *AK056164* (2.5 kb), *AF429305* (1.1 kb), and *AF429306* (1.1 kb) (Figure 1A). In the adult mouse, expression of the 2 kb *Rmst* isoform is largely restricted to the CNS (Uhde et al., 2010). Similarly, in humans, *RMST* exhibits a brain-specific expression (Ng et al., 2012). *RMST* expression increases during neuronal differentiation of hESCs (Figures S1A–S1B available online) and is abundant in neurons differentiated from H9 and ReN-VM cell lines (C_T value of *RMST* ~20; C_T value of *ACTB* ~16) (Figure 1E). *RMST* is also a stable RNA transcript that can be detected by northern blot. The two bands at approximately 2.5 kb and 1.1 kb correspond to the alternatively spliced isoforms of *RMST* (Figure S1C). The stronger 2.5 kb signal indicates that *AK056164* is the major *RMST* transcript in neurons.

Nuclear lncRNA *RMST* Regulates Neuronal Differentiation and Associates with Transcription Factor SOX2

RMST orthologs from human to frog are well conserved at important gene regulatory regions, including their promoter regions, first exons, and splice sites (Chodroff et al., 2010). Coupled with a brain-specific expression of *RMST* (Ng et al., 2012; Uhde et al., 2010), this implies that *RMST* has important roles in brain development and/or function. As such, we examined the roles of *RMST* by means of knockdown and overexpression of the lncRNA in neural progenitors. Knockdown of *RMST* in ReN-VM NSCs and H9-derived neural progenitors by siRNAs prevented neuronal differentiation (Figure 2A), resulting in cells alternatively adopting a glia fate (Ng et al., 2012; Figure 2B). Conversely, overexpression of the major *RMST* isoform



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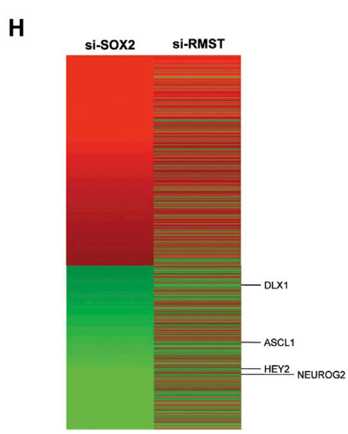
	si-RMST	si-SOX2	overlap
differentially expressed	1171	626	626
up-regulated	632	372	331
down-regulated	539	254	152



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Top 10 GO terms for the 152 genes downregulated upon SOX2 and *RMST* knockdown

	Gene Ontology Biological Process	Gene Ontology Term	*p*-value
1	Neuron migration	GO:0001764	5.28E-06
2	Regulation of Notch signaling pathway	GO:0008593	6.55E-06
3	Cellular response to stress	GO:0033554	2.97E-05
4	Forebrain development	GO:0030900	2.97E-05
5	Neurogenesis	GO:0022008	1.29E-04
6	Positive regulation of Notch signaling pathway	GO:0045747	1.46E-04
7	Telencephalon development	GO:0021537	1.48E-04
8	Rho protein signal transduction	GO:0007266	1.67E-04
9	Brain development	GO:0007420	1.81E-04
10	Oligodendrocyte differentiation	GO:0048709	1.89E-04

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(*AK056164*) in human neural progenitors resulted in increased neuronal marker expression and a larger percentage of TUJ1-expressing neurons, indicating the importance of *RMST* in neuronal differentiation (Figures 2C–2E).

Having established that *RMST* has an important function in neuronal differentiation, we sought mechanistic insight into the role of *RMST* in neurogenesis. Because miR-135A2 and miR-1251 are transcribed from the same genomic locus as *RMST* (Figure 1A), we questioned whether *RMST* could be a precursor transcript for these miRNAs or regulate their expression. To this end, we performed a knockdown of *RMST* in ReN-VM NSCs and H9-derived NPCs and measured microRNA (miRNA) levels by quantitative PCR (qPCR) 48 hr after transfection of siRNAs. We found that knockdown of *RMST* had no effect on miR135A2 and miR1251 levels in both ReN-VM NSCs and H9-derived NPCs (Figure S2A), indicating that *RMST* did not alter the expression levels of its neighboring miRNAs and that the effects on neurogenesis were not due to changes in these miRNAs.

The localization of *RMST* within the cell was assessed, given that this could provide clues to the possible interacting partners of *RMST*. Subcellular fractionation experiments had established that *RMST* is predominantly a nuclear transcript (Ng et al., 2012). Using RNA fluorescence in situ hybridization (RNA-FISH), we confirmed the nuclear localization of *RMST* (Figure 3A). This suggests that *RMST* functions in the nucleus and could interact with nuclear proteins and/or chromatin.

In a search for possible protein partners of *RMST*, RNA pull-down experiments were performed as previously described (Huarte et al., 2010). In brief, biotin-labeled RNA was transcribed in vitro, immobilized on streptavidin magnetic beads and then incubated with nuclear extract prepared from ReN-VM neurons. Complexes captured on the beads were subjected to stringent washes and finally eluted for SDS-PAGE. Unique bands that were present in the *RMST* pull-down and not in the beads only (no RNA) or antisense-*RMST* pull-downs were excised for protein identification by mass spectrometry (Figures 3B and S2C). Two specific interacting protein partners of *RMST* were identified and confirmed by western blot: hnRNPA2/B1, an RNA-binding protein, and SOX2, a transcription factor (Figure 3C). Given that hnRNP-K has been previously shown to bind lncRNAs (Huarte et al., 2010), we asked whether hnRNP-K could also interact with *RMST*. The inability of biotin-labeled *RMST* to pull down hnRNP-K and BRN2, a SOX2-interacting transcription factor, indicated that *RMST*'s interactions with hnRNPA2/B1 and SOX2 are unique and specific (Figure 3C). Consequently, to confirm the biochemical interactions of *RMST* with these two proteins, we performed crosslinked RNA immunoprecipitation (RIP) on the ReN-VM neurons and detected the enrichment of *RMST* in both SOX2 and hnRNPA2-FLAG pull-downs (Figure 3D). Altogether, this set of RNA-protein interaction assays confirmed that *RMST* binds specifically to SOX2 and hnRNPA2/B1 both in vitro and in vivo.

RMST and SOX2 Regulate Common Target Genes

Given that depleting *RMST* led to loss of neuronal differentiation, we questioned whether the depletion of *RMST*-interacting proteins would also result in a similar phenotype. ReN-VM NSCs and H9-derived NPCs were transfected with siRNAs targeting *RMST*, hnRNPA2/B1, and SOX2, and, subsequently, differentiation was induced in N2B27 media. Because efficient SOX2 knockdown during neuronal differentiation resulted in massive cellular death (Cimadamore et al., 2011), a less-potent SOX2 siRNA was used for these experiments (Figure 3F). The extent of neuronal differentiation was measured 1 week later by immunostaining for the presence of neuronal markers TUJ1 and MAP2. Loss of neuronal differentiation was seen upon knockdown of hnRNPA2/B1 or SOX2, a phenotype similar to the depletion of *RMST* (Figures 3E and S2).

We subsequently focused on the specific interaction between *RMST* and SOX2, given that hnRNPA2/B1 are ubiquitous RNA-binding proteins, whereas SOX2 plays a pivotal role throughout neural development. To investigate gene regulation by SOX2 and *RMST*, we downregulated SOX2 and *RMST* in differentiating ReN-VM cells and performed a microarray analysis 48 hr after siRNA transfection. In this microarray experiment, data were normalized to the nontarget siRNA control data set. A large overlap of differentially expressed genes between the si-RMST and si-SOX2 data sets indicated that *RMST* and SOX2 regulate a common set of target genes (Figure 3G; Tables S1–S2). Notably, all the differentially expressed genes in SOX2 knockdown NSCs were also misregulated in *RMST* knockdown NSCs, often in the same direction (Figure 3H), indicating that gene regulation by

Figure 3. Nuclear lncRNA *RMST* Associates with Nuclear Proteins hnRNPA2/B1 and SOX2 and Coregulates SOX2 Target Genes

(A) RNA fluorescent in situ hybridization (RNA-FISH) was performed with a specific fluorescent probe targeting *RMST*. Colocalization of the *RMST* signal and the DAPI nuclear stain indicated that *RMST* is a nuclear lncRNA.

(B) Schematic representation of the RNA pull-down experiment for identifying specific proteins interacting with *RMST*.

(C) After biotinylated RNA pull-down, the eluate was resolved on SDS-PAGE for western blotting. Antisense RMST (AS-RMST) and "no RNA" (beads only) samples served as negative controls. Bands corresponding to hnRNPA2/B1 and SOX2 were only observed in RMST pull-down. hnRNP-K, an RNA-binding protein known to interact with lncRNAs, and BRN2, a partner transcription factor of SOX2, were not detected in the RMST pull-down.

(D) Formaldehyde-crosslinked RNA immunoprecipitation (RIP) was carried out, confirming the in vivo interaction of *RMST* with hnRNPA2 and SOX2 ($p = 0.002$ and $p = 0.02$, respectively). Error bars are SD.

(E) A nontarget siRNA (si-NT) along with siRNAs targeting RMST (si-RMST), hnRNPA2/B1 (si-hnRNPA2B1), and SOX2 (si-SOX2) were introduced into ReN-VM NSCs, and their ability to differentiate into neurons was determined by immunostaining for TUJ1- and MAP2-expressing cells. Representative images from each condition are shown.

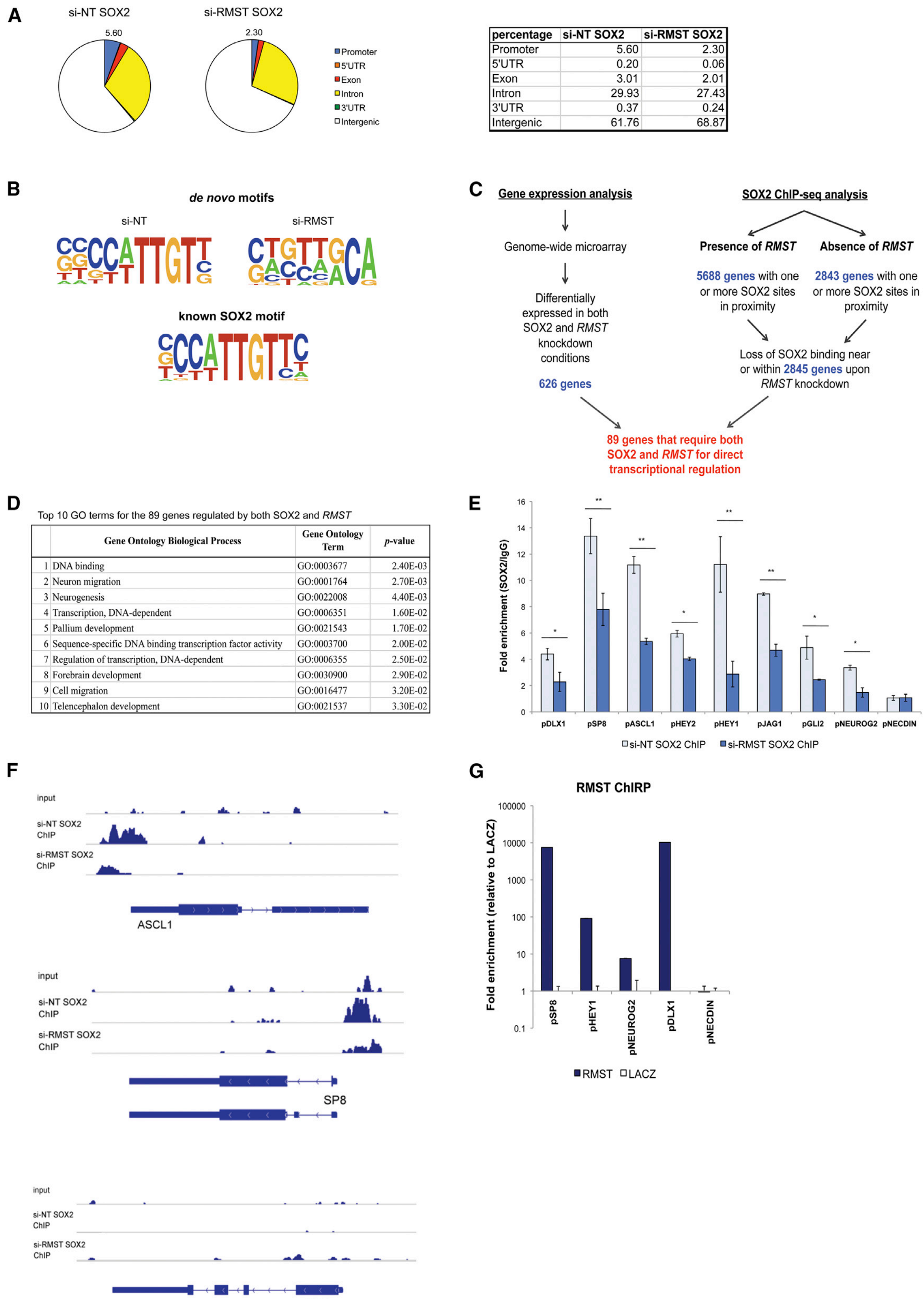
(F) RNAi efficiencies of the siRNAs used. We chose a less efficient SOX2 siRNA for this study because NSCs with insufficient SOX2 undergo cell death. Error bars are SD; p values are as indicated.

(G) Summary of the microarray results.

(H) Differentially expressed or misregulated genes upon knockdown of SOX2 and *RMST* are shown. The number of genes in the overlap is depicted.

(I) Top ten gene ontology (GO) terms of the 152 genes that are downregulated in the si-SOX2 and si-RMST overlap.

See also Figure S2.



percentage	si-NT SOX2	si-RMST SOX2
Promoter	5.60	2.30
5'UTR	0.20	0.06
Exon	3.01	2.01
Intron	29.93	27.43
3'UTR	0.37	0.24
Intergenic	61.76	68.87

	Gene Ontology Biological Process	Gene Ontology Term	*p*-value
1	DNA binding	GO:0003677	2.40E-03
2	Neuron migration	GO:0001764	2.70E-03
3	Neurogenesis	GO:0022008	4.40E-03
4	Transcription, DNA-dependent	GO:0006351	1.60E-02
5	Pallium development	GO:0021543	1.70E-02
6	Sequence-specific DNA binding transcription factor activity	GO:0003700	2.00E-02
7	Regulation of transcription, DNA-dependent	GO:0006355	2.50E-02
8	Forebrain development	GO:0030900	2.90E-02
9	Cell migration	GO:0016477	3.20E-02
10	Telencephalon development	GO:0021537	3.30E-02

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SOX2 and *RMST* are correlated. The list of genes coactivated by both *RMST* and SOX2 include *ASCL1*, *NEUROG2*, *HEY2*, and *DLX1* (Figure 3H), which are neurogenic genes, supporting the roles of *RMST* and SOX2 in neurogenesis (Cimadamore et al., 2011; Ferri et al., 2004; Puligilla et al., 2010). In addition, this group of 152 *RMST* and SOX2-activated genes showed an enrichment of gene ontology (GO) terms related to neurogenesis and neuronal function (Figure 3I).

Although *RMST* and SOX2 share a large set of common targets, *RMST* did not appear to regulate SOX2 expression. Knockdown of *RMST* did not alter SOX2 expression at the transcript level. Conversely, knockdown of SOX2 had no effect on *RMST* levels (Figure S2B). This indicated that *RMST* does not regulate downstream target gene expression by acting upstream of SOX2 and vice versa.

SOX2 Binding Is Affected in the Absence of *RMST*

Next, we investigated whether *RMST* could be a transcriptional coactivator of SOX2. Through ChIP sequencing (ChIP-seq) experiments, we determined the genome-wide binding profiles of SOX2 in the presence and absence of *RMST*. If *RMST* affects the transcription factor activity of SOX2, then we would expect an altered chromatin-binding profile of SOX2 upon *RMST* knockdown. In the presence of *RMST* (si-NT SOX2 ChIP library), 55,771 total SOX2 binding sites were observed, and 21,329 peaks were observed in the proximity of genes. However, upon *RMST* knockdown (si-RMST SOX2 ChIP library), only 13,575 peaks of 43,601 SOX2 sites were near genes, suggesting that almost half of all SOX2 occupancy near genes was lost upon *RMST* knockdown. In particular, more than half of the SOX2 binding sites in the promoter (−1 kb to +1 kb) region were lost when *RMST* was depleted (Figure 4A), suggesting that *RMST* is required for SOX2 binding at some promoters.

Using all SOX2 peaks for further analysis, we found that, in the presence of *RMST*, the SOX2-occupied sites demonstrated strong enrichment of the SOX2 consensus sequence motif ($p = 1 \times 10^{-901}$), indicating that these are genuine SOX2 binding sites (Figure 4B). Notably, when *RMST* was depleted from cells, the SOX2 motif appears degenerate (Figure 4B). SOX2 binding in *RMST* knockdown conditions also appears to have shifted, suggesting that *RMST* is required for proper SOX2 DNA-binding activity. This altered SOX2 binding pattern, along with the observation that less SOX2 binds to chromatin in the absence of *RMST*, indicates that *RMST* is required for SOX2 binding at certain genomic loci.

RMST and SOX2 Bind at Promoter Regions of Neurogenic Genes and Coregulate Their Expression

We were interested in identifying genes that are directly regulated by both SOX2 and *RMST*. To achieve this, the SOX2 ChIP-seq data were integrated with global gene expression data collected by microarray analysis of cells depleted of *RMST* and SOX2 expression (Figure 4C). SOX2 binding sites, which are present in the control, but not when *RMST* is depleted, implies the requirement of *RMST* for chromatin binding. In control cells, we found 21,329 SOX2 binding sites near genes, and these binding sites were assigned to 5,688 known genes. When *RMST* was depleted, SOX2 binding sites at 2,845 of these genes were lost, indicating the requirement of *RMST* for SOX2 binding at these loci. Then, this set of 2,845 genes was overlapped with the list of 626 genes that were differentially expressed upon knockdown of SOX2 and *RMST*, producing a list of 89 genes that are directly transcriptionally regulated by SOX2 and *RMST* (Table S3). GO analysis of these 89 genes revealed an overrepresentation of terms related to nervous system development and transcriptional regulation (Figure 4D). This suggests that *RMST* is required for SOX2 binding at regulatory regions of neurogenic transcription factors. Thus, we measured SOX2 occupancy at promoter regions of known neurogenic markers and SOX2 targets (Engelen et al., 2011) by ChIP-qPCR. As expected, many of the promoter target regions showed decreased SOX2 occupancy upon *RMST* knockdown (Figure 4E). This was confirmed by the ChIP-seq tracks, where SOX2 displayed decreased peak height upon *RMST* knockdown at the promoter regions of *ASCL1* and *SP8* (Figure 4F).

The observation that *RMST* was essential for SOX2 binding at the promoters of neurogenic genes prompted us to investigate whether the lncRNA binds to chromatin at those sites as well. Using a recently developed technique for probing lncRNA-chromatin interactions (Chu et al., 2011), we were able to isolate *RMST*-bound chromatin fragments by chromatin isolation by RNA purification (ChIRP). ChIRP-PCR revealed strong *RMST* occupancy at the promoter regions of neurogenic transcription factors such as *SP8*, *HEY1*, *NEUROG2*, and *DLX1*, consistent with our prior observation that *RMST* is required for SOX2 binding at these promoter regions (Figures 4G and S3). This confirms

Figure 4. *RMST* Partners with SOX2 and Binds to SOX2-Bound Chromatin In Order to Regulate Gene Expression

(A) Pie chart of the distribution of SOX2 binding sites in the presence (si-NT SOX2) and absence (si-RMST SOX2) of *RMST*. The percentage of sites located within promoter regions (−1 kb to +1 kb) is shown on the pie chart, whereas the percentages of sites corresponding to other genomic loci are summarized in the adjacent table.

(B) De novo motifs generated from HOMER 4.1 software with all binding sites from the respective SOX2 ChIP libraries. The TRANSFAC SOX2 motif is also shown.

(C) A flowchart illustrating the number of genes that require both *RMST* and SOX2 for direct transcriptional regulation by overlapping ChIP-seq and microarray data.

(D) GO analysis of the 89 genes that require both *RMST* and SOX2 for transcriptional regulation. The top ten most significant GO terms show an overrepresentation of terms relating to transcriptional regulation and nervous system development.

(E) The extent of SOX2 binding at promoter regions of the indicated genes in the presence (si-NT SOX2 ChIP) or absence of *RMST* (si-RMST SOX2 ChIP). pNECDIN was a negative control region. Error bars are SD. A Student's t test was performed to test for significance. $^{*}p < 0.05$; $^{**}p < 0.01$.

(F) Depiction of the SOX2 binding profile in the upstream regions of *ASCL1*, *SP8*, and *NEFL*. The genomic region upstream of *ASCL1* and *SP8* showed reduced SOX2 binding upon knockdown of *RMST*. NEFL, which is not a SOX2 target gene, does not exhibit any binding in its upstream region.

(G) The extent of *RMST* association with chromatin was detected by ChIRP followed by qPCR. Binding of *RMST* (RMST ChIRP) to the promoter regions of the indicated genes was compared relative to a control (LACZ ChIRP). pNECDIN served as a negative control region. Error bars are SD.

See also Figure S3.

that *RMST* has DNA-binding properties and functions as a plausible transcriptional coregulator of SOX2.

Altogether, our data indicate that *RMST* is an important RNA component in the neural regulatory network. In NSCs where *RMST* is not expressed, SOX2 binds to and activates its target genes in order to maintain NSC identity. During neuronal differentiation, the downregulation of REST leads to increased *RMST* expression, which binds to SOX2 as well as chromatin. In doing so, it activates the transcription of neurogenic genes such as *ASCL1* and *DLX1*, which drives the neuronal differentiation pathway.

DISCUSSION

In the current study, we performed an in-depth analysis of the molecular and cellular functions of *RMST*, an lncRNA that modulates neurogenesis. Through the use of genome-wide ChIP-seq, microarray, and ChIRP approaches, we established that *RMST* is a transcriptionally regulated lncRNA that associates with SOX2 in order to regulate downstream genes and has an indispensable role in neuronal differentiation. Our genome-wide gene perturbation results provide strong evidence that *RMST* participates in the global regulation of gene expression. More than 1,000 genes were differentially expressed upon *RMST* knockdown, which was comparable to the knockdown of transcription factors, and this reaffirms that *RMST* is a regulatory molecule. We also show that *RMST* binds to the promoter regions of some SOX2 target genes, promotes SOX2 binding at these regions, and modulates the transcription of these genes. SOX2 binding in the absence of *RMST* was not completely abrogated at the promoter regions studied (Figures 4E–4F), which is consistent with our findings that a modest SOX2 knockdown ($\sim$40% to 50%) was sufficient to affect neurogenesis without causing cell death. Altogether, these results imply that *RMST* behaves as a transcriptional coactivator of SOX2. To date, only a few lncRNAs are known to be transcriptional coregulators, *Evf-2* being the most well-characterized example. *Evf-2* associates with and acts as a cofactor for the transcription factor Dlx2 to regulate transcription of the neighboring *Dlx5* and *Dlx6* genes (Feng et al., 2006). The finding that *RMST* associates with SOX2 to regulate downstream targets also suggest that the cofactor lncRNA mechanism might be more common than expected.

Although SOX2 is known to maintain NSC identity (Graham et al., 2003), it is also required for neuronal differentiation (Cimadamore et al., 2011; Ferri et al., 2004; Puligilla et al., 2010). We saw that the knockdown of SOX2 prevented neuronal differentiation (Figures 3E and S2), but overexpression of SOX2 did not enhance neurogenesis (Figure S4), consistent with reported findings. It is not unusual for transcription factors such as SOX2, which has multiple protein partners, to have seemingly opposing cellular functions. For instance, the master transcription factor OCT4 can interact with SOX2 or SOX17 to maintain pluripotency or initiate endodermal differentiation, respectively, in ESCs (Aksoy et al., 2013; Stefanovic et al., 2009). SOX2 could also have similar interaction-partner-dependent roles in the decision of neural cell fate.

Interestingly, our results suggest that SOX2 possesses RNA-binding capabilities, a fact that underlies the mechanistic action of *RMST*. However, it is still unclear whether SOX2 associates with *RMST* directly or through SOX2-interacting proteins that bind to *RMST*. Either way, it is clear from this study that SOX2 and *RMST* exist in a transcriptional complex during neuronal differentiation and that they regulate many common downstream targets that are important for this process.

Although we focused mainly on the SOX2-*RMST* interaction throughout this study, the interaction between *RMST* and hnRNPA2/B1 is also specific and robust. The RNA-binding proteins hnRNPA2/B1 have important cellular functions. They regulate alternative splicing in cells (David et al., 2010; Kamma et al., 1999) and facilitate mRNA trafficking to neuronal dendrites (Shan et al., 2003), and the loss of hnRNPA2/B1 activity is also associated with neurodegeneration (Kim et al., 2013; Sofola et al., 2007). Hence, the interaction between *RMST* and hnRNPA2/B1 may indicate the participation of the lncRNA in alternative splicing, mRNA trafficking, or neuronal cell survival. Although these are speculative and subject to experimental validation, future studies could potentially uncover an additional layer of gene regulatory complexity conferred by *RMST*.

In conclusion, we provide mechanistic insight into the molecular function of *RMST*, an lncRNA we found to be necessary for neuronal differentiation. *RMST* has all the hallmarks of a functional lncRNA: (1) *RMST* is specifically expressed in the brain tissues, (2) *RMST* is transcriptionally regulated, (3) *RMST* exhibits some degree of conservation across species, and (4) perturbation of *RMST* levels results in a phenotype that matches its molecular function. In a broader sense, this study has important implications for gene regulatory circuits that govern cell-fate determination. In combination with transcription factors, lncRNAs can provide an additional layer of complexity to gene regulatory networks that control cellular differentiation. In the case of neurogenesis, the lncRNA *RMST* acts as a transcriptional coregulator of SOX2.

EXPERIMENTAL PROCEDURES

Neural Stem Cell Culture

The human NSC line, ReN-VM was maintained on Matrigel (BD Biosciences) in NSC media containing epidermal growth factor and basic fibroblast growth factor (both from Invitrogen) as previously described (Donato et al., 2007). Differentiation of ReN-VM cells was initiated 1 day after passaging by replacing NSC medium with N2B27 media (Ying et al., 2003). For neuronal differentiation, ReN-VM cells were exposed to the N2B27 media for at least 7 days.

Neural Differentiation of hESCs

The hESC line H9 (WiCell) was maintained on Matrigel in mTeSR media (STEMCELL Technologies) and serially passaged with dispase. Neural induction of undifferentiated H9 cells was achieved by dual SMAD inhibition (Chambers et al., 2009) with 1 μM dorsomorphin and 10 μM SB431542 (both from Stemgent) in N2B27 media for 10 days. For neuronal differentiation, the H9-derived neural progenitors were exposed to N2B27 media supplemented with 20 ng/ml brain-derived neurotrophic factor and 20 ng/ml glial cell-line-derived neurotrophic factor.

Expression Analyses

For expression analyses by microarray, total RNA was isolated in experimental triplicates from NSCs 48 hr after transfection with the respective siRNAs. Total RNA was converted into biotin-labeled single-stranded DNA with an Illumina TotalPrep RNA Amplification Kit (Applied Biosystems) and hybridized to HumanHT-12 v4 Expression BeadChip (Illumina) according to the manufacturer's recommendations. Array data quality control, normalization, and statistical analysis were as described previously (Ng et al., 2012). GO analyses were performed with DAVID v6.7.

qPCR analyses on complementary DNA (cDNA) transcribed with a High-Capacity cDNA Reverse Transcription Kit was performed on an ABI 7900HT Fast Real-Time PCR System (both from Applied Biosystems) and normalized to *GAPDH* or *U1* expression. A list of primers is provided in Table S4.

RNA Interference

siRNA sequences were designed with Invitrogen's Block-It RNAi Designer (https://rnaidesigner.invitrogen.com/rnaiexpress/). At least two duplexes were designed for each gene, and the most effective siRNAs were used for subsequent studies. The sequences of the duplexes are provided in Table S5. For transfection, ReN-VM cells were seeded at a density of 2×10^5 cells per well of a six-well plate. The respective siRNAs were complexed with Lipofectamine RNAiMAX (Invitrogen) according to the manufacturer's instructions. For prolonged knockdown, a second transfection was repeated 3 days after the initial transfection.

Overexpression of *RMST*

Full-length *RMST* and control LACZ RNA were transcribed in vitro with T7 RNA Polymerase (New England Biolabs) followed by purification of the RNA with RNeasy MinElute columns (QIAGEN). The molecular weight of the RNA was checked by agarose gel electrophoresis, and the RNA was folded by adding an equal volume of RNA structure buffer (20 mM Tris [pH 7.0], 0.2 M KCl, and 20 mM MgCl2), heated to 70°C for 5 min, and slowly cooled to room temperature in order to allow proper secondary structure formation. Then, the full-length in vitro transcribed RNA was transfected into ReN-VM cells with TransIT-mRNA transfection (Mirus Bio). Transfection efficiency was determined by means of a Cy3-labeled *RMST* RNA, which showed about 75% transfection.

Immunofluorescence

Cells were fixed in 4% paraformaldehyde (PFA), permeabilized with 0.1% Triton X-100 for 15 min, blocked in a solution comprised of 5% fetal bovine serum and 1% BSA in PBS, and incubated overnight at 4°C with the following antibodies: rabbit anti-TUJ1 (Covance, 1:2,000), mouse anti-MAP2 (Abcam, 1:500), mouse anti-S100β (Sigma-Aldrich, 1:1,000), and rabbit anti-GFAP (Dako, 1:500). Secondary antibodies conjugated with AlexaFluor-488 or AlexaFluor-594 (Molecular Probes and Invitrogen) were diluted 1:2,000 in blocking buffer and incubated for 2 hr at room temperature. DAPI (0.5 μg/ml) was used to visualize cell nuclei. Digital images were captured on a Zeiss Axio Observer.

RNA Fluorescent In Situ Hybridization

A fluorescence probe labeling *RMST* was prepared with a full-length *RMST* cDNA with a Prime-It Fluor Fluorescence Labeling Kit (Stratagene) according the manufacturer's protocol. Neuronal cells were seeded onto Matrigel-coated glass slides 1 day prior to RNA-FISH. Then, cells were rinsed once in PBS, extracted in cytoskeleton (CSK) buffer (100 mM NaCl, 300 mM sucrose, 3 mM $MgCl_2$, and 10 mM PIPES) at room temperature, incubated in CSK and detergent buffer (CSK buffer, 0.5% Triton X-100, and 10 mM Vanadyl Ribonucleoside Complex), fixed in 4% PFA, and dehydrated following serial incubations in 70%, 80%, 95%, and 100% ethanol. Fluorescent probes were heat denatured and incubated with the cells at 37°C overnight in a humidified chamber protected from light. Then, slides were washed with a series of saline sodium citrate buffers the next day, and cellular DNA was counterstained with DAPI.

Biotinylated RNA Pull-Down

The RNA pull-down assay was performed as previously described (Tsai et al., 2010). In brief, in vitro transcription of RNA was performed with Biotin RNA Labeling Mix (Roche) and T7 RNA polymerase. RNA was folded in RNA structure buffer (as described above) and adsorbed onto streptavidin magnetic beads (NE Biolabs). ReN-VM neuronal cell lysate was prepared by briefly sonicating 10 million cells in RIP buffer (150 mM KCl, 25 mM Tris [pH 7.4], 0.5 mM DTT, 0.5% NP-40, and complete protease inhibitors). Then, the RNA-loaded beads were incubated with precleared cell lysate at 4°C for 4 hr. Next, beads were washed five times in RIP buffer and eluted in Laemmli buffer, and the retrieved proteins were run on SDS-PAGE gels for mass spectrometry or western blot. The dilutions for western blot of the following primary antibodies are as follows: goat anti-SOX2 (Santa Cruz Biotechnology, 1:200) and mouse anti-hnRNPA2/B1 (Sigma-Aldrich, 1:1000).

RNA Immunoprecipitation

RIP was performed as previously described (Ng et al., 2012). For each RIP assay, 5 μg of antibodies was used. Given that an optimal hnRNPA2/B1 antibody was not available for RIP, mouse anti-FLAG (Sigma-Aldrich) was used to pull down RNA associated with hnRNPA2-FLAG that was ectopically expressed in ReN-VM cells. For SOX2 RIP, rabbit anti-SOX2 (ab59776 from Abcam) was used.

Chromatin Immunoprecipitation

For the REST ChIP experiment, ReN-VM cells were crosslinked with 1% formaldehyde, and the reaction was quenched by the addition of 125 mM glycine. Then, the cells were washed and lysed with cell lysis buffer, consisting of 50 mM HEPES (pH 7.5), 1 mM EDTA (pH 8.0), 150 mM NaCl, 0.1% sodium deoxycholate, 0.1% SDS, 1% Triton X-100, and complete protease inhibitors (Roche). Next, chromatin was sheared to fragments of 100–500 bp by sonication for seven cycles on high amplitude (cycles of 30 s on followed by 30 s off) with the Bioruptor. Then, 5 μg of rabbit anti-NRSF or rabbit IgG isotype control antibody were adsorbed onto protein G magnetic beads and incubated with chromatin extract at 4°C overnight. Crosslinking of DNA fragments was reversed by pronase and subsequent incubation at 42°C for 2 hr and 68°C for 8 hr. Recovered DNA was subject to RNase treatment and analyzed via qPCR.

For the SOX2 ChIP assay, ReN-VM cells were transfected with the respective siRNAs and induced in order to differentiate in N2B27 medium. The efficiency of knockdown was assessed 48 hr after differentiation induction by qPCR, and cells were treated as above for ChIP. We used 5 μg of goat anti-SOX2 or goat IgG isotype control antibody. Recovered DNA was analyzed by qPCR and sent for high-throughput sequencing on an Illumina Genome Analyzer IIx. Relative occupancy values or fold enrichment values were calculated by determining the relative immunoprecipitation efficiency (ratio of the amount of immunoprecipitated DNA to that of the input sample) and normalized to the level observed at the control region. Sequences of primers used for the ChIP-qPCR quantification are provided in Table S6.

ChIP-Seq Analysis

Sequencing reads were aligned with the human genome assembly hg18 as the reference genome. For the analysis of ChIP-seq results, we applied the MACS 1.4.2 software with the default parameters for peak finding. In brief, redundant reads that could result from the overamplification of ChIP DNA were removed, and peak enrichment was calculated relative to the genome background. A threshold of $p = 10^{-5}$ was used to call significant peaks. An "input" sample was also included for eliminating nonrandom enrichment (Feng et al., 2012). To filter for high-confidence ChIP-seq peaks, we plotted the difference between the SOX2 libraries and input mapped reads against random SOX2 peaks. Then, we selected for peaks that had a greater peak height in comparison to random peaks. For motif discovery, the HOMER 4.1 algorithm (Heinz et al., 2010) was applied to each set of ChIP-seq libraries for de novo binding motif discovery and known motif analysis with default parameters. All significant peaks were used for motif analysis in both sets of SOX2 ChIP-seq experiments. The HOMER 4.1 software was also used for annotating the peaks on the basis of their overlap with the hg18 genome assembly RefSeq genes transcription start site (by default defined from −1 kb to +100 bp), transcription termination site (by default defined from −100 bp to +1 kb), exons, 5′ UTR, 3′ UTR, introns, and intergenic regions.

Chromatin Isolation by RNA Purification

ChIRP was performed as previously described (Chu et al., 2011). Differentiating ReN-VM cells in N2B27 medium were crosslinked with 1% glutaldehyde, and the reaction was quenched in 125 mM glycine. The cell pellet was resuspended in nuclear lysis buffer consisting of 50 mM Tris-Cl (pH 7.0), 10 mM EDTA, 1% SDS, protease inhibitors, and RNase inhibitors and subjected to lysis by sonicating for 30 cycles on the Bioruptor. Chromatin was hybridized with the respective biotin-labeled probe sets and incubated at 37°C for 4 hr. Then, the biotin-probe-chromatin complexes were pulled down with streptavidin magnetic beads, washed, eluted by proteinase K, and subjected to either DNase I or RNase A treatment. Recovered RNA and DNA were analyzed by qPCR. The sequences for the biotin-labeled probes are provided in Table S7.

ACCESSION NUMBERS

The NCBI GEO accession number for the ChIP-seq data reported in this paper is GSE49405.

SUPPLEMENTAL INFORMATION

Supplemental Information contains four figures and seven tables and can be found with this article online at http://dx.doi.org/10.1016/j.molcel.2013.07.017.

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